U5 Summary

"This pdf covers most of the details."

Have fun with this gorgeous syllabus 😍😇

Made by: Nour

TOPIC 8

Neurons

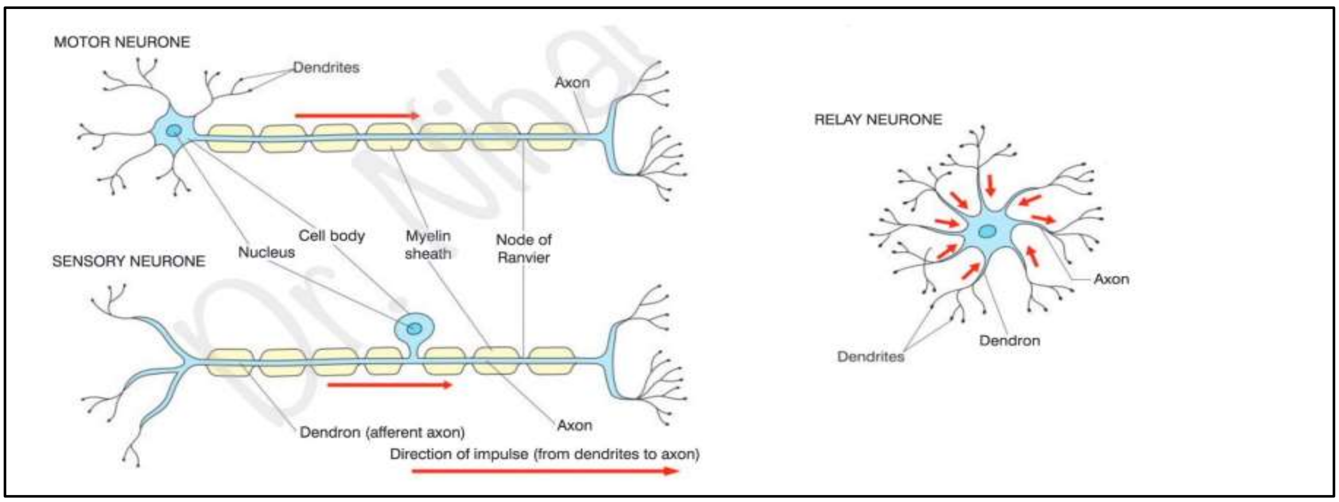
➢ Structure of nerve cells:

1- Dendron: Extension from cell body sub divided into dendrites carry nerve impulses towards the cell body.
2- Axon: Long fibre carrying nerve impulses away from the cell body.
3- Schwann cells: Wrap around the axon creating myelin sheath. Providing insulation to the axon allowing saltatory conduction.
4- Nodes of Ranvier: The space between adjacent Schwann cells which speeds up rate of conduction.

✓ Spinal reflex like hand withdrawal:

1- Stimulus ...heat object detected by receptors.
2- Receptor ...if temperature receptors detect temperature above the threshold value this will generate an action potential.
3- Action potential pass along the sensory neuron to spinal cord.
4- Relay neurone link the sensory to motor neurone via the synapse.
5- Motor neurone carry action potential to the effector (biceps muscles).
6- Muscle contract ..bring a response which is raising up the hand away from hot object.

➢ Types of neurons:



Sensory neurone	Relay neurone	Motor neurone
Dendron is longer	Short dendrites	Many Shorter dendrites
Dendron is myelinated	X	X
Shorter axon Myelinated axon	Shorter axon X	Long axon Myelinated axon
Central cell body / towards the middle	- - - - - -	Terminal cell body

- Properties of the membrane of axon:

1- Axon cell membrane is made of phospholipid bilayer that restrict ion movement.
2- Yet it has sodium potassium pump which transports sodium and potassium ions across the membrane.
3- Has voltage gated sodium and potassium channels, they alternate between being open and close to allow diffusion of ions.

- Nerve impulse:

- Electrical event produced by charge differences between the outside and inside of the neuron membrane.
- It is based on ion movements through specialized protein pores and by an active pumping mechanism.

1- Neuron cell membranes are polarized at rest: (resting potential)

- P.d= -70 mV.
- Sodium potassium pump, Na+ actively transported out the axon and potassium ion in using ATP.
- High K+ ion concentration inside the axon and high Na+ ion concentration outside axon.
- Sodium ion channels are closed so can't diffuse back again into the axon.
- The membrane of neurone is permeable to K+ so they will diffuse out of the axon down its concentration gradient.
- Making outside of the membrane is positively charged compared to the inside.
- Electrical gradient will pull potassium back into the axon.

2- Neuron cell membrane become depolarized: (action potential)

- P.d= -50 mV…+40 mV.
- A stimulus triggers ion channels called voltage gated sodium ion channels to open.
- So the permeability of the neurone membrane to NA+ increases.
- So sodium ions diffuse into the neuron down sodium ion electrochemical gradient.
- Causing inside to become more positive causing depolarization until it reaches +40 mV.

3- Repolarisation:

- At this point Voltage gated Na+ ion channels will close so sodium ions can't continue to enter the neurone.
- Where some of the potassium K+ VOLTAGE gated channels open.
- K+ outflux down its electrochemical gradient and concentration gradient.
- So inside of axon will become negative relative to outside.

4- Hyperpolarization:

- P.d below -70 mV.
- Membrane potential becomes more negative than resting level (polarization).
- This occurs due to prolonged opening of voltage gated potassium ion channels.
- Causing excess potassium ions to diffuse out of the neuron.
- So the axon is more negative inside than outside.

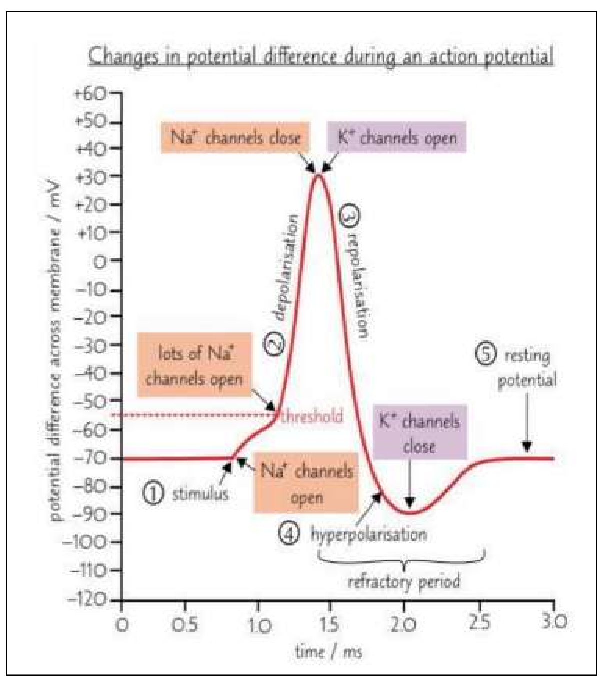
Refractory period:

- After an action potential, the neuron cell membrane can't be excited again straight away.
- This is because the ion channels are recovering and they can't be made to open.
- This ensures that action potentials travel in one direction.

✓ Importance of refractory period:

1- It controls the frequency of impulses conducted by the neuron.

2- Regulates the timing of an action potential.



➢ Action potential is an All or Nothing law:

- There is a certain level of stimulus, (threshold) that triggers the impulse.
- If the potential difference is below the threshold no impulse generated.

> ✓ Why resting potential requires energy?
> - Because ATP is used by sodium-potassium pumps.
> - To move Na+ and K+ across the cell membrane.
> - To establish concentration gradient for sodium and potassium ions.

➢ The speed of conduction of nerve impulses depends on:

1- Diameter of the axon:

- As diameter increases, the resistance to flow decreases. Therefore, as diameter of axon increases speed of conduction increases.

2- Myelination:

- Allow depolarisation only at nodes of Ranvier.
- So the action potential will jump from one node to node in a process called Saltatory conduction.
- Allowing faster conduction of action potential.

➢ Why the speed of conduction in a myelinated neuron is faster than in a non-myelinated neuron?

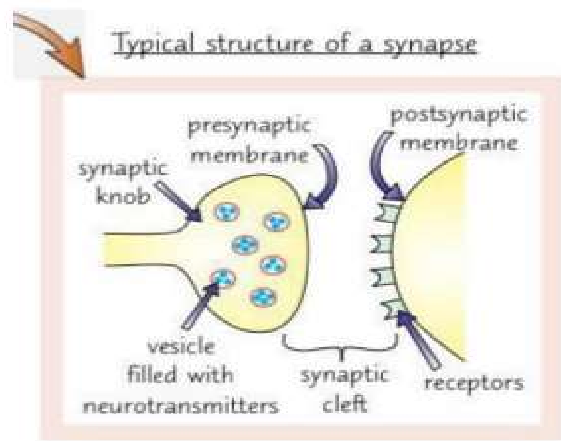
- Myelin sheath insulates the axon.
- There're breaks in the myelin sheath of myelinated fibers.
- Action potential occur at nodes of Ranvier.
- So the action potential will jump from one node to node in a process called Saltatory conduction.

Synapse and neurotransmitters

- A synapse is a junction between a neuron and the next cell.
- The tiny gap between the cells at a synapse is called the synaptic cleft.

✓ Why impulses travel in one direction across a synapse?

- Neurotransmitters/vesicles are only found in presynaptic knob.
- Receptors for neurotransmitters are found only on postsynaptic



➢ Role of synapse:

1- Transport information between neurones as synapse is found between neurones.
2- Ensure one way transmission.
3- Protection of effectors (muscles and glands) from over stimulation.

➢ Transmission of nerve impulse across a synapse:

1- An action potential arrives at the synaptic knob of presynaptic neuron.
2- The action potential stimulates voltage gated calcium ion channels in the presynaptic neuron to open.
3- Calcium ions diffuse into the synaptic knob.
4- This causes the synaptic vesicles to fuse with the presynaptic membrane.
5- The vesicles release the neurotransmitter into the synaptic cleft by exocytosis.
6- The neurotransmitter diffuses across the synaptic cleft and binds to specific receptors on the postsynaptic membrane.
7- This causes voltage gated sodium ion channels in the postsynaptic neuron to open.
8- The influx of sodium ions into the postsynaptic membrane causes depolarization and an action potential on the postsynaptic membrane is generated.
9- The neurotransmitter is removed from receptors so the response doesn't keep happening. They are taken back into the presynaptic neuron or broken down by enzymes.

Neurotransmitter:

- Neurotransmitters are needed to transmit the nerve impulse across the synapse.
- Released from vesicles in the presynaptic neurone/ knob.
- Diffuse across the synaptic cleft and bind to receptors on the post synaptic neurone.
- Initiating the action potential in the post synaptic neurone.

Effect of drugs on the nervous system:

1- Nicotine: Highly addictive

- It mimics the effect of acetylcholine and binds to specific acetylcholine receptors (nicotine receptors) in post synaptic membrane.
- Trigger an action potential in the post synaptic neurone but then receptors remain unresponsive to more stimulation for some time.
- Nicotine cause increase in heart rate and blood pressure.
- Trigger release of another neurotransmitter (dopamine) for pleasure sensation.

2- Lidocaine:

- Block the VG sodium channels so no influx of sodium ions into the neurone.... no depolarisation.... no neurotransmitters are released in next synapse... no impulse reaching the brain.

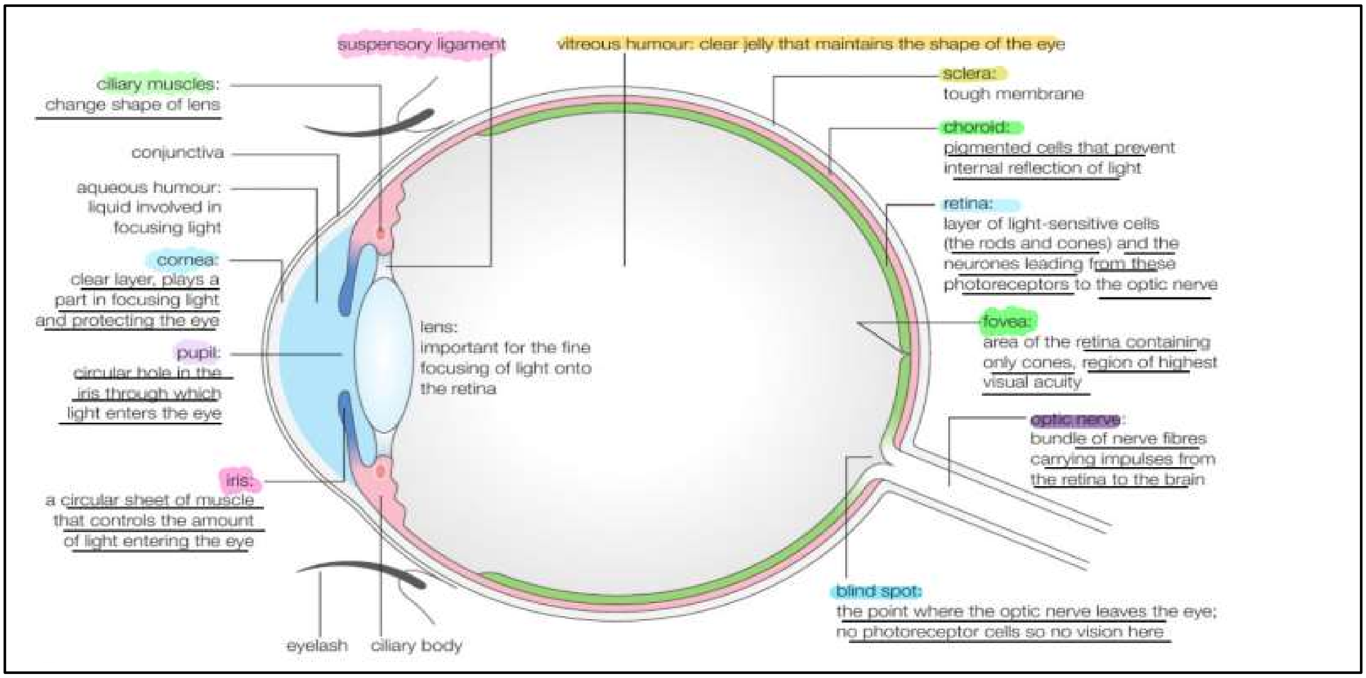
A) Prevent you from feeling pain

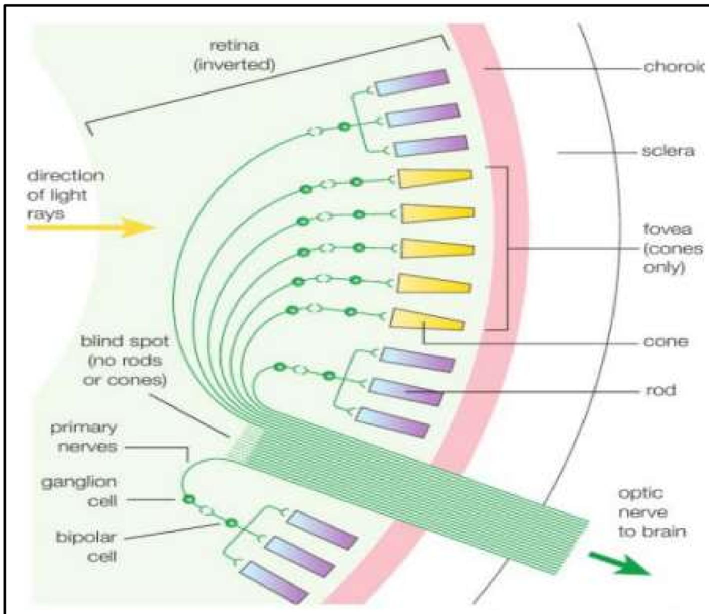
B) use to prevent some heart arrythmias (irregular heart beat) ....block sodium channels increasing depolarization threshold ....this reduces early or extra action potentials from pacemaker.

3- Cobra venom (alpha cobra toxin):

- Binds reversibly to acetylcholine receptors in post synaptic membrane and neuromuscular junctions.
- Thus, prevent transmission of impulses across synapses (neuromuscular junctions).
- So muscles not stimulated to contract and gradually causes paralyses.
- In case toxins reach muscles involved in breathing it causes death.

The human eye





Structure of the retina:

The Retina is formed of 3 layers

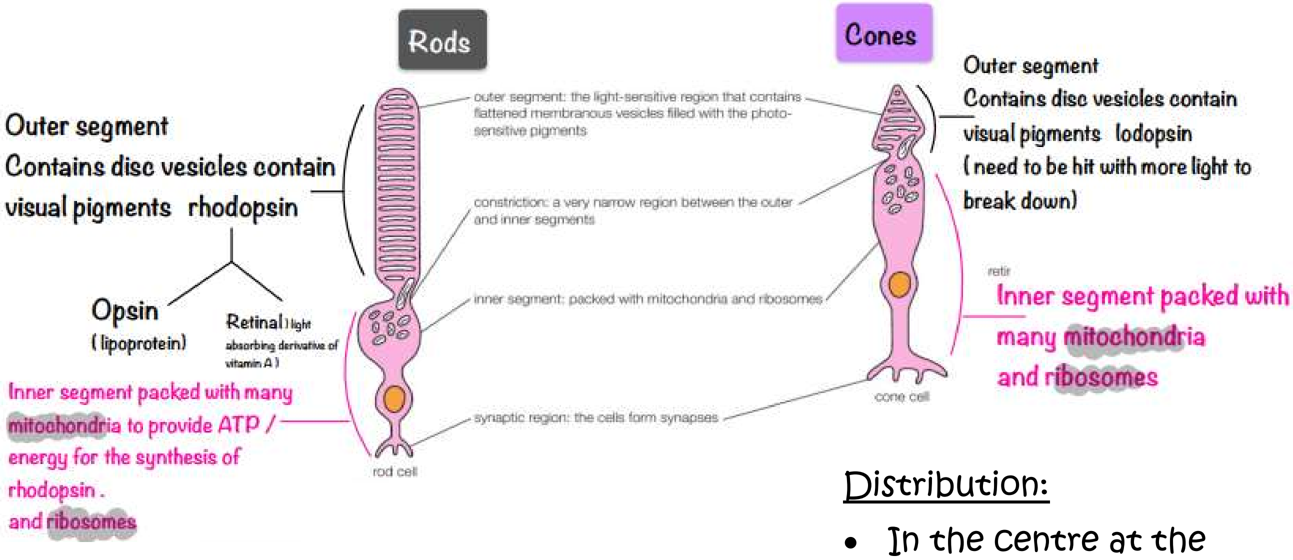
Outermost: Layer of rods and cones (photoreceptors).

Middle: Bipolar cells, they are called so because they synapse with 2 layers.

Innermost: Ganglion cells give rise to axons that form fibres of the optic nerve.

- Photoreceptors are light receptors in your eye

- Light enters the eye through pupil. The amount of light that enters is controlled by the iris (circular and radial muscles).
- Light rays are focused by the lens onto the retina. The retina contains photoreceptors cells that detect light.
- Fovea is an area of the retina where there are lots of photoreceptors.
- Nerve impulses from the photoreceptor cells are carried from the retina to the brain by the optic nerve, which is a bundle of neurons.
- Where the optic nerve leaves the eye is called the blind spot. There aren't any photoreceptor cells, so it's not sensitive to light.



Distribution:

- Peripheral parts of retina.
- Not found at the fovea.

Type of pigments:

- Rhodopsin (opsin + retinal).
- Respond to low light intensity giving black and white vision.

Distribution:

- In the centre at the fovea.

Type of pigments:

- Iodopsin.
- Respond to bright light and give clarity of colour vision.

➢ When light falls on the retina (Light adaptation):

- rod cells are stimulated
- In rods the visual pigment is rhodopsin which absorbs light.
- Light energy causes cis-retinal to be converted to trans-retinal.
- This causes rhodopsin to break into trans-retinal and opsin (bleaching).
- Bleaching of rhodopsin causes the sodium ion channels to close.
- So sodium ions are actively pumped out of the cell but they can't diffuse back.
- Inside of the membrane much more negative than the outside so the cell membrane is hyperpolarized.
- When the rod cell is hyperpolarized, it stops releasing neurotransmitters. This means there's no inhibition of the bipolar neuron.

➢ When it's dark, (Dark adaptation):

- your rods aren't stimulated
- Trans-retinal is converted to Cis-retinal.
- Cis retinal joins to the opsin again forming rhodopsin using energy.
- Sodium ions are pumped out of the cell using active transport.
- But sodium ions diffuse back into the cell through sodium channels.
- This makes the inside of the cell negative compared to the outside so cell membrane is depolarized.
- This triggers the release of neurotransmitters.
- But the neurotransmitters inhibit the bipolar neuron – the bipolar neuron can't fire an action potential so no information goes to the brain.

➢ Pupil reflex:

- Circular muscles are involved.
- They are antagonistic muscles and work opposite to each other.
- Stimulus received by sensory receptors in retina in the eye.
- So action potential occurs where reflex arc is formed.
- Impulse passes through sensory neuron in optic nerve to rely neuron in brain.
- Then pass across the motor neuron which is connected to muscles (radial or circular) of iris.

➢ Bright light:

- Circular muscles contract and radial muscles relax
- Pupil reflex is the reflex contraction and relaxation of the antagonistic muscle of the iris.
- Therefore, pupil becomes smaller/ constrict so less light enters the eye.

➢ DIM light:

- Circular muscles relax and radial muscles contract.
- Pupil reflex is the reflex contraction and relaxation of the antagonistic muscle of the iris.
- Therefore, pupil becomes larger/dilates so more light enters the eye.

✓ Importance of ATP in rods?

- ATP provides energy
- To convert Trans retinal into Cis retinal
- And re-joins retinal to the opsin again forming rhodopsin.

Habituation: It is a form of learning involving the decreased responsiveness to a repeated stimulus.

That is not reinforced by reward or punishment.

➢ **Mechanism of habituation:**

1- Decreased responsiveness of the presynaptic neurone to calcium ions / calcium ion channels don't open.

2- Less neurotransmitters fuse with presynaptic membrane.

3- Less neurotransmitter releases from the presynaptic membrane.

4- Less binding to receptors on the post synaptic membrane.

5- Less depolarisation of the postsynaptic neurone.

6- Less action potential generated in the postsynaptic neurone (motor).

➢ **Importance/ advantages of habituation:**

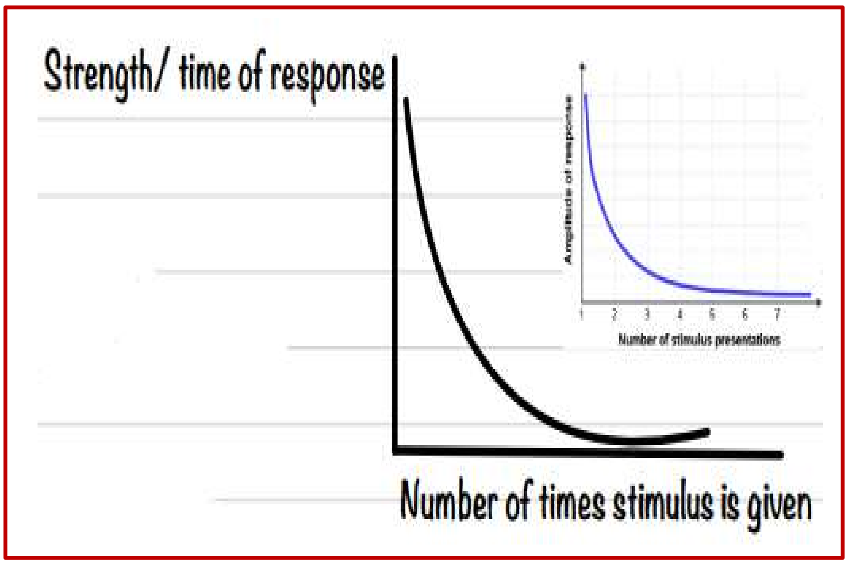
- Stimulus is ignored since its harmless/ not threatening.
- Where less withdrawal of siphons save energy for other activities.

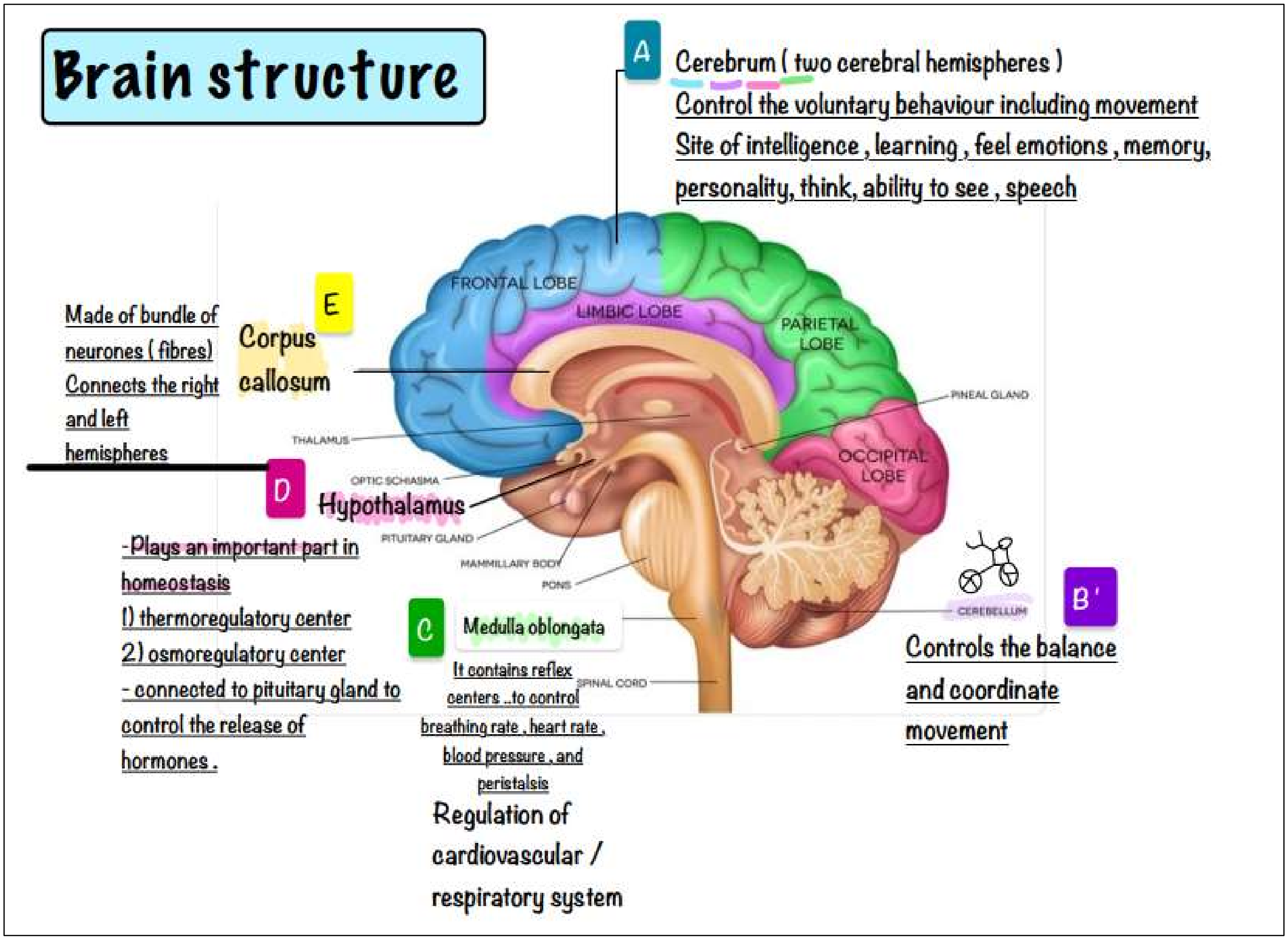
➢ **Factors affecting how fast habituation takes place:**

1- Frequency of the stimulus they are exposed to

2- Duration of stimulation

3- Strength of the stimulus





Nervous system

1- CNS (central nervous system):
- Brain
- Spinal cord

2- PNS (peripheral nervous system):
- Sensory neuron
- Motor neuron
  - Somatic nervous system
    - Voluntary nervous system under the conscious control.
  - Autonomic nervous system
    Not under conscious control
    - Control of heart rate, breathing rate and constriction of iris/ vessels.

Sympathetic-nervous system:

- Neurotransmitter is noradrenaline.
- Increase heart rate and breathing rate.
- Dilates the eye pupil.

Para sympathetic nervous system:

- Neurotransmitter is acetylcholine.
- Decrease heart rate and breathing rate.
- Constrict the pupil.

Brain imaging

➢ **Importance of brain imaging:**

1. Detect any abnormality in brain (example bleeding, density change).

2. Detecting the type, size, location of abnormalities (tumours, oedema, stroke, haemorrhage).

3. Follow up effective of a treatment by repeated scans and comparisons.

4. Looking for surgical accessibility (according to site / size).

✓ Why abnormalities in different parts of brain cause different symptoms:

1. different regions of brain affected.

2. Different areas of brain have different functions.

3. Symptoms depends on region of brain affected.

4. Where different type of abnormalities causes different symptoms.

1- CT (computed tomography):

- CT can only identify larger/ main structures.
- CT scan gives location/ size.
- CT gives 2D/ 3D images.
- Cheap and available.

➢ Disadvantages:

- CT is not safe due to the use of X-rays.
- Low resolution.
- Does not allow observation of the brain in action.

2- MRI (magnetic resonance imaging):

- MRI uses radio waves/ magnetic waves.
- MRI has a higher resolution.
- Gives 2D/ 3D images.
- More details seen.
- Safe because no use of X-rays.

➢ Disadvantages:

- MRI is expensive.
- MRI is noisy, need to keep still.

3- fMRI (functional magnetic resonance imaging):

- fMRI scan shows the activity of brain directly.
- fMRI measures uptake of oxygen.
- Active area of brain gets more blood so more oxygen.
- Active areas have more oxyhemoglobin.
  - Supplying oxygen to active cells.
- More active areas appear light/ white.
- fMRI can allow brain activity to be seen in real time.
- Safe as does not use X-rays.

✓ Why a region of brain might appear lighter in an image by fMRI?

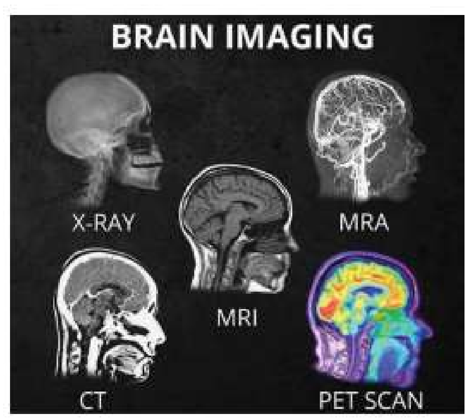
- Due to more activity.
- An increase in oxygenated blood/ blood flow to this region.
- fMRI signals reflected/ not absorbed by oxygenated blood.

4- PET scan (positron emission tomography):

- Patient takes an isotope that emits positrons/ is incorporated into glucose.
- More active neurons with increased respiration.
- Will require increased supply of glucose.
- Positrons emitted from glucose produce gamma rays that detected and converted into an image.

➢ Advantages:

- Provides information about the function.
- PET scan shows areas that are more metabolically active.
- Provides a 3D image.



Chemical balance of the brain

Communication between the neurons around the body and in brain depend on a delicate balance of naturally occurring chemicals.

An imbalance in these transmitters can result in both mental and physical symptoms.

- ➢ The blood brain barrier:

1- The capillaries in brain have very tight narrow endothelium (narrow pores).
2- This protects the brain from many pathogens in blood as it makes it difficult for the bacteria to cross into brain and cause infection.

➢ Parkinson's disease "affects body movement"

Mechanism of disease:

Gradual damage/ degeneration to nerve cells in Substantia Nigra which normally responsible for the release of dopamine.

✓ Why the reduction in dopamine by presynaptic neurones affect the motor action?

- Fewer dopamine molecules to bind to the receptors / ligand gated sodium channels.
- On the post synaptic membrane.
- Initiating fewer action potential in post synaptic neurone.
- So fewer impulses sent to parts to brain controlling the motor function / muscles.

➢ Symptoms:

Slowness of movement.

Stuffiness of muscles.

Poor balance.

Difficulty in walking.

➢ Treatment of Parkinson's disease

1- L Dopa:

- Can across the blood brain barrier.
- L Dopa is converted to dopamine in the brain.
- Dopamine binds to receptors on the post synaptic membrane.

2- Monoaminoxidase B/ MAOB: (An enzyme inhibitor)

- To prevent the breakdown of dopamine.
- Dopamine binds to receptors on the post synaptic membrane.

3- Dopamine agonists:

- Dopamine agonist can cross the blood brain barrier.
- It binds to receptors on the post synaptic membrane.

Depression

Decreased level of serotonin (neurotransmitter).

Symptoms:

- unable to enjoy life.
- No self-esteem.
- Loss of appetite.
- Thinking of committing suicide.

Treatment of depression:

1- SSRIs (selective serotonin reuptake inhibitor):

- They block the re uptake channels of serotonin in the presynaptic membrane.
- More serotonin in synaptic cleft.
- More binding to the receptors on the post synaptic membrane.
- More action potentials in post synaptic neuron.

2- TCA (tricyclic antidepressant):

- Increasing level of serotonin and nor adrenaline in brain.
- They block reuptake channels of Serotonin and Noradrenaline.

3- MDMA (Ecstasy): methylenedioxy-N-methylamphetamine

- Blocks reuptake of Serotonin.
- So that the synapse is completely flooded with serotonin.
- Giving a feeling of happiness, elation, confidence and inflation.

However, it is an illegal drug due to the following side effects:

- Affect hypothalamus.
- Hypertension.
- Tachycardia (increase in heart beats).
- Arrhythmia.
- Increased ADH secretion leading to oliguria.

Coordination systems in plants

- Phototropism

A) Light is coming from one side.

- Auxins to be redistributed in the shaded sides stimulate the cell elongation.
- Bending towards light.

B) light is coming from both sides.

- Auxins are equally distributed on both sides so grow straight up.

C) if no light.

- Grow straight up (more vertical growth).

- Phototropism... Auxins (IAA)

- Auxins are secreted from the tip of the shoot, shoot tip act as photoreceptors to light.
- The auxin diffuses down wards (shaded side) stimulating cell elongation.
- Auxins bind to receptor in the cell membrane.
- Auxin stimulate proton pumps in the cell membrane to actively transport hydrogen ions from the cytoplasm into the space in the cell wall.
- Cell wall become acidic (decrease in pH).
- Activate proteins and weaken the cell wall, breaking hydrogen bonds between cellulose microfibrils and dissolve pectin…. loosening cell wall.
- Stimulating the potassium ions to enter the cells decreasing the water potential.
- So water will enter by osmosis causing an increase in turgor pressure.
- Causing cell walls to stretch allowing parts to grow in size /causing cell elongation.
- So shoot grow towards light.

Gibberellin

- Water is absorbed by the seed.
- Stimulate the production of gibberellin by the embryo within the seed.
- Stimulate the production of amylase.
- Amylase diffuse and move to the endosperm catalyse the breakdown of starch into maltose.
- Then maltase stimulates the breakdown of maltose into glucose.

✓ How gibberellin causes an increase in the length of plants?

- Gibberellin enters the nucleus/ binds to the surface receptors.
- Gibberellin act as a transcription factors/ secondary messenger activating protein synthesis making mRNA.
- Causing cell elongation.

Photoreceptors phytochrome

Light receptors in the plant known as phytochrome.

There are two main types of light:

1- Red light: bright light / Day time
2- Far Red Light: moon light / Dark / Night

A) Phytochrome Red (PR) …..(P660) absorb red light , is an inactive form

B) Phytochrome Far Red (PFR) ….(P730) absorb far red light is physiologically active .

Effect of phytochrome on germination:

Far red light (another light source) is applied to seeds.

PFR (active phytochrome) is responsible for germination of certain types of seeds.

• This means that REd lights stimulate germination, while far red light inhibits germination.

• The one exposed to red light …Pr is converted into active Pfr (active phytochrome) where PFR stimulate germination.

• And the Pr inhibits the germination.

• Yet the effect of light is reversible.

Length of the time it takes for one form of pigment to be converted into another depends on light intensity

In low (in dark) Takes minutes

Pfr to be converted to Pr SLOWLY , but no Pr is converted back

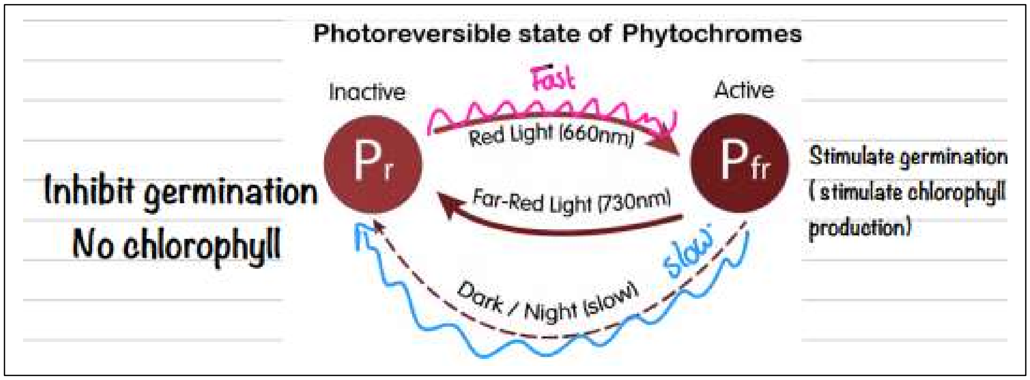
Pfr stimulate the germination / Pr inhibit germination

Level of Pfr become too low to stimulate germination / level of Pr becomes too high to inhibit the germonation

Pr is a more stable form of pigment .. but its the Pfr that is biologically active

In light Takes few seconds

The higher the light intensity the more the Pfr produced at a faster rate



✓ Why the percentage of seeds that germinate in no light gradually decrease?

- PFR slowly changes to PR in dark.
- Levels of PFR become too low to stimulate germination.
- Seeds become less viable with storage (substrate used up).

Genetic engineering

Genetically modified organism: Organisms that have genome (DNA) altered by genetic engineering.

Genetic engineering: Taking a gene from one organism and inserting it into another organism.

- The process of making a protein using genetic engineering has a number of stages:

Stage 1: Obtain/ synthesis the desired gene

Cut the gene of the desired protein (synthesis of the desired gene) by restriction enzyme leaving recognition sites.

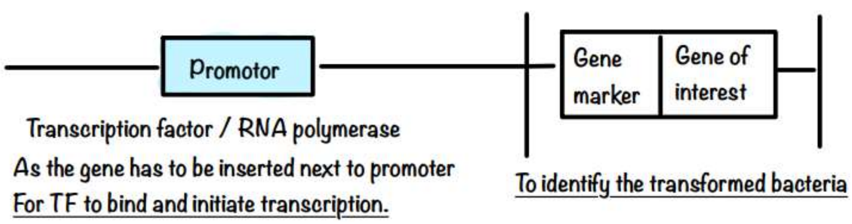
Step 2: Make many copies of the desired gene using PCR using Taq DNA polymerase.

Step 3: Vector which is the plasmid removed from the bacteria and cut open using the same restriction enzyme.

Leaving sticky ends which has complementary base sequence to sticky ends of the gene.

Stage 4: Insert the gene and seal it using DNA ligase enzyme, this is a recombinant DNA because it has DNA from 2 different species.

Stage 5: Add a marker gene to the plasmid with recombinant DNA. This marker gene is usually a gene of antibiotic resistance.



Stage 6: Then insert the plasmid into the host cell (bacteria) using a gene gun and culture the bacteria on a medium containing a specific antibiotic so that only marked bacteria will grow while other bacteria die.

Advantages of using bacteria in genetic engineering:

Medical benefits:

- Lower risk of rejection as they are produced by human DNA so they are human proteins recognized by the immune system as self and they don't trigger an immune response.
- Large scale production.

Agricultural benefits:

Increase quality and quantity of food by:

- Inserting genes for pest resistance, this improves the crop yield with no need for pesticides.
- Inserting genes for herbicide resistance to limit the usage of herbicides.

Evaluate use of viruses in genetic engineering:

- Viruses are hollow so can carry genes.
- Viruses attach to specific cells and their genetic material enters the cell.

Disadvantages:

- Viruses have antigens than bind to specific receptors on their target cells and therefore cannot attach to all cell types.
- Some viruses are too small to carry all the genetic material inside them.
- Some viruses may cause disease.
- People may be immune to a particular virus.

Hazards of Genetic Engineering

1. In humans:

- It opens doors for genetic manipulation of sperms and ova prior to fertilization (designer babies) which is ethically arguable.
- May have unpredictable long-term outcomes like cancers.

2. In crops:

- The genes of herbicide resistance may escape to other plants through cross pollination.
- GM crops may turn into super weeds.

3. In animals:

- The release of GE animals in the wild may cause disturbances of their food chains and consequently their habitat.
- May be used to develop destructive biological weapons.

Microarrays

A technique used …….in two ways:

1. Comparing the gene present in complete genomes of two different species (gene analysis).

2. Finding out which genes are expressed in tissues or cells at any given time (gene expression profiling).

- Using microarray technique to identify genes that are actively transcribed…

- The mRNA from two types of cells (diseased and undiseased) is extracted and collected ….sample collection and isolation of mRNA.
- Then use reverse transcriptase to use mRNA as template to produce cDNA
- Then cDNA is labelled with fluorescent tags (denatured to give single stranded DNA) ……creation of labelled cDNA.
- Then allowed to hybridise with probe on microarray.
- Each probe is unique to a different gene.
- Spots on the microarray that fluoresce indicate the genes that were transcribed in the cell… (indicating genes expressed).
- Collection and analysis using bioinformatics.
- Compare the sequence of the active genes from large number of individuals with and without disease to identify mutation.
- The intensity of light emitted by each spot indicate the level of activity of each gene ….high intensity indicates that many mRNA were present in the sample and vice versa (light intensity is proportional to degree of expression)
- Thus knowing activity of gene.

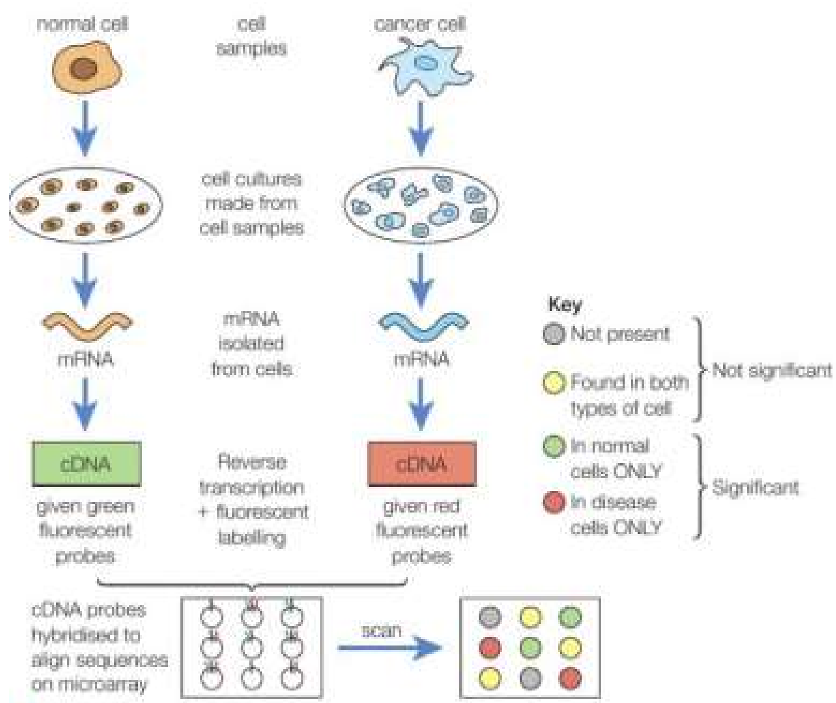


fig A Using a DNA microarray.

Bioinformatics

It's the development of the software and computing tools that stores , organise and analyses data about the genomes (DNA sequences) of living organisms.

- **Bioinformatics is aiming for:**

1- Facilitate access to large amount of data.
2- Facilitate access to data on DNA and proteins.
3- Large data base .....where we Use data base to find probes.
4- Fast, accurate ......using computer softwares and statistical analysis.

Using bioinformatics, we can process and use the information generated using microarrays and other forms of DNA analysis to adapt the care of individual patients and learn more about the biology of many different organisms.

Drug testing

1- Preclinical phase which involves testing on animals or human cells to see how it works in whole organisms.

2- Clinical phase I, Drug being tested on healthy volunteers.

3- Then review by independent scientists to see if work can progress for phase 2.

4- Clinical phase II, tested on small number of patients (500 to 1000) using appropriate concentration/ dosage.

5- Clinical phase III, drug is being tested on patients with larger group over 5000 to monitor the effectiveness and safety.

6- (Placebo and double-blind trial are used for both phase II and III).

7- Where a number of patients should be placed randomly into two groups.

A) One group will receive the drug containing chemical compound

B) And the other will receive the placebo, acting as a control group taking the drug without active ingredient for comparison.

8- Double blind, where neither the doctor nor the patients know who is taking the real drug to remove bias.

9- Statistical analysis and test for significant differences.

- <u>Using animals in medical research raises ethical issues</u>

Arguments AGAINST using animals in medical research:

- Animals are different from humans, so drugs tested on animals may have different effects on humans.
- Experiments can cause pain and distress to animals.
- There are alternatives to using animals in research, e.g. Using cultures of human cells or using computer models to predict the effect of experiments
- Some people think that animals have the right to not be experimented on, e.g. animal rights activists.

Arguments FOR using animals in medical research:

- Animals are similar to humans, so research has led to loads of medical breakthroughs, e.g. antibiotics, insulin for diabetics and organ transplants
- Animal experiments are only done when it's absolutely necessary and scientists follow strict rules, e.g. animals must properly looked after, painkillers and anaesthetics must be used to minimize pain
- Using animals is currently the only way to study how a drug affect the whole body- cell cultures and computers aren't a true representation of how cells may react when surrounded by other body tissues. It's also the only way to study behaviour.
- Other people think that humans have a great right to life than animals because we have more complex brains, e.g. compared to rats, fish, fruit flies (which are commonly used in experiments).